

**Designed pair:** reachable, hazard-free, genuinely emitting  
 $\text{core}(t_1) = \text{core}(t_2)$ ,  $\text{gen}(t_1) = \text{gen}(t_2)$ ;  $p_2 = (c_2, e_2)$

**state  $t_1$**   
 $\text{pay}(E_1) = [p_1, p_2]$   
 $p_2$  already emitted

**state  $t_2$**   
 $\text{pay}(E_2) = [p_1]$   
 $p_2$  still owed at  $c_2$

for every  $\text{obs}$  with  $\text{obs}(t_1) = \text{obs}(t_2)$ ,  
every  $P = g \circ \text{obs}$  has  $P(t_1) = P(t_2) = B$

yes

$p_2 \in B?$

no

**state  $t'_1$**   
 $p_2$  appears twice  
 $\text{duplicate}(t'_1)$   
 $\Rightarrow \text{unsafe}(t'_1)$

**state  $t'_2$**   
 $p_2$  remains absent  
 $\Rightarrow \text{lost}(c_2, t'_2)$

**Either way, both repaired stores are clean at  $c_2$ :**  
 $\text{clean}(t'_1, c_2) \wedge \text{clean}(t'_2, c_2)$